Exp No: 28
COUNTING SEMAPHORE IN SYSTEMC

AIM:
To implement Counting Semaphore allowing multiple resource access.

SOFTWARE REQUIRED:
- Ubuntu (or WSL Ubuntu)
- g++
- CMake
- SystemC

PROGRAM:

```
#include <systemc.h>

SC_MODULE(counting_semaphore) {
   int sem;

   void task() {
      while(true) {
         wait(1, SC_SEC);

         if(sem > 0) {
            sem--;

            cout << "Task Accessing Resource at "
                 << sc_time_stamp()
                 << " Remaining: " << sem << endl;

            wait(2, SC_SEC);

            sem++;
         }
      }
   }

   SC_CTOR(counting_semaphore) {
      sem = 2;

      SC_THREAD(task);
      SC_THREAD(task);
      SC_THREAD(task);
   }
```

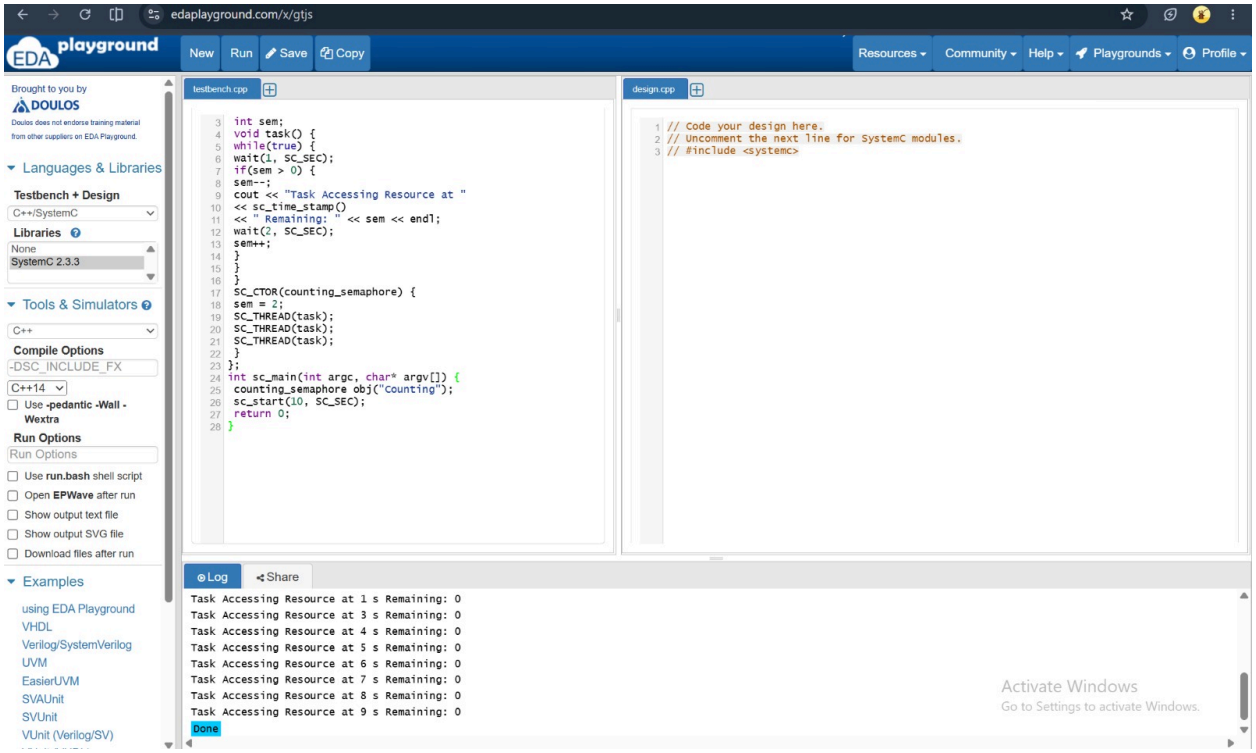
```
};

int sc_main(int argc, char* argv[]) {
    counting_semaphore obj("Counting");

    sc_start(10, SC_SEC);

    return 0;
}
```

OUTPUT



RESULT

Thus the Counting Semaphore was created using SystemC and verified.